

Exam 1 Blueprint Guide for Students

Exam 1 – 30 questions – 45 minutes

****This list is not exhaustive, but it will help guide the major concepts of this unit**

1. Pharmacology Foundations (~20%)

You should be able to:

- Differentiate among major branches of pharmacology
- Explain how medications produce effects in the body
- Explain how receptor activity influences medication effects
- Differentiate between agonists and antagonists
- Explain factors that influence medication response

2. Pharmacokinetics and Medication Concentrations (~30%)

You should be able to:

- Explain absorption, distribution, metabolism, and excretion (ADME)
- Identify major organs involved in metabolism and excretion
- Explain half-life
- Explain first-pass metabolism
- Explain concepts related to medication concentrations and therapeutic effects

3. Medication Response Concepts (~15–20%)

You should be able to:

- Differentiate among common medication responses
- Explain how medications may interact
- Explain tolerance, dependence, therapeutic effect, adverse effect, synergism and why they occur

4. Medication Safety and Nursing Responsibilities (~20–25%)

You should be able to:

- Explain the nurse's role in medication safety
- Identify required medication order components
- Recognize incomplete medication orders
- Explain medication reconciliation
- Describe nursing responsibilities related to controlled substances
- Explain the importance of medication education

5. Autonomic Nervous System Concepts (~10–15%)

You should be able to:

- Compare sympathetic and parasympathetic responses
- Explain the effects of cholinergic medications
- Explain the effects of anticholinergic medications
- Explain sympathetic stimulation

6. CAM, OTC Medications, Alternative Therapies (<5%)

You should be able to:

- Describe basic safety considerations related to herbal, over the counter, and alternative therapies
- Explain why these products are important during medication reconciliation

You should be able to:

Define major pharmacology concepts in your own words

Compare closely related concepts (e.g., pharmacodynamics vs pharmacokinetics)

Explain how medications move through the body

Explain how medications produce effects

Explain why medication responses differ among individuals

Identify medication safety principles and nursing responsibilities

Apply pharmacology concepts to simple scenarios